

# CONTROL ENGINEERING III EEC 433

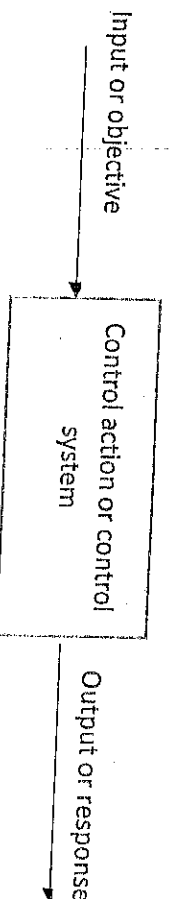
## 1.0 INTRODUCTION

A control system is an interconnection of components forming a system configuration in which any quantity of interest is maintained or altered in accordance with a defined manner. The basic parts of control system can be described by:

1. Input or objective of control.
2. Control action or control system,
3. Output or response.

### 1.1 Basic control system

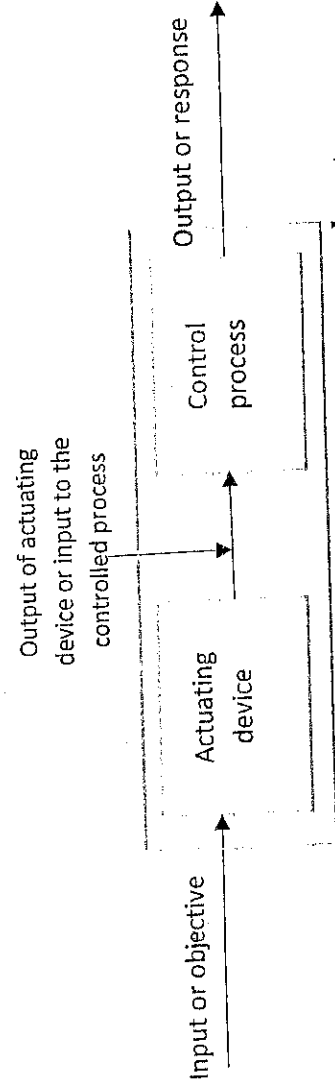
Basic control system is as shown in fig 1.1



*Fig. 1.1 basic control system*

### 1.2 Open loop control system

Without feedback or non feedback control system is known as open loop control system. The element of an open loop control system can usually be divided into two parts: the actuating device and the controlled process. As shown in Fig. 2 the actuating device may be an amplifier, actuator, mechanical linkage, etc depending on the system.



**Fig 1.2 Elements of open loop control system**

An open loop control system utilizes an actuating device to control the process directly that means the output has no effect (or control) on the control action.

Because of the simplicity and economy of open-loop control system, it finds application in many non critical systems. Basically this type of systems are economical but usually inaccurate.

Examples of open loop control systems are;

1. Bread toaster,
2. Electric heater
3. Automatic washing machine.

### 1.2.1 Advantages of open loop systems

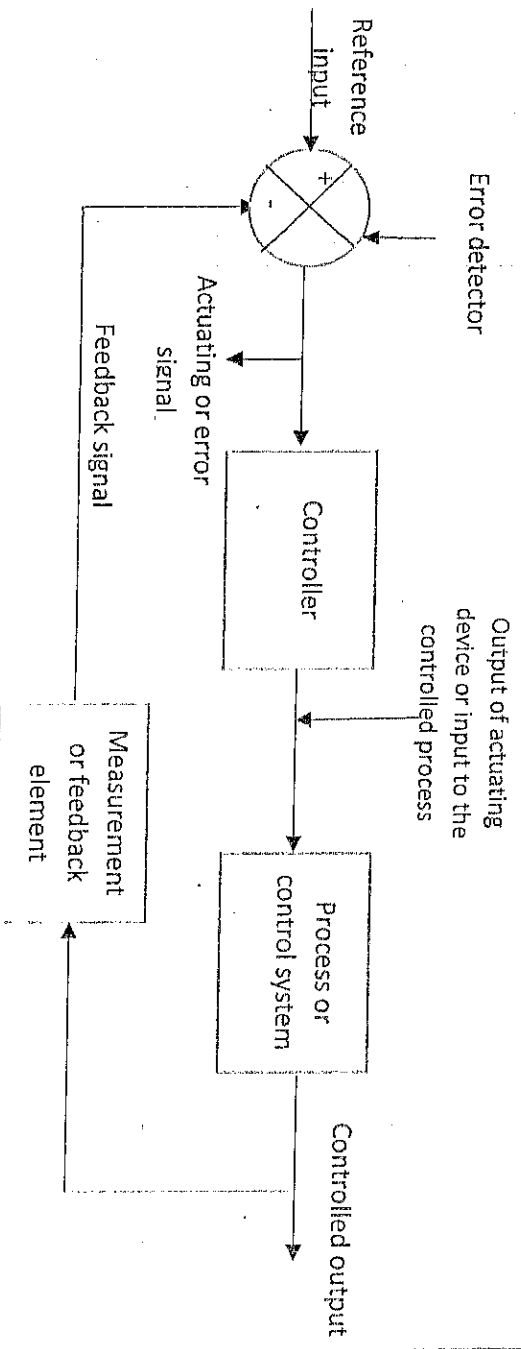
1. Simple in construction
2. They are cheap
3. They are generally stable
4. Require less maintenance
5. Calibration is easy

### 1.2.2 Disadvantages of open-loop systems

1. They are inaccurate, accuracy depends on calibration point.
2. They are not reliable, operation is affected by the non linearities in the elements.
3. Optimization of open loop system not possible.

### 1.3 Closed loop control system

Feedback control system is known as closed-loop control system. It maintains a prescribed relationship between the output and the reference input by comparing them and using the difference (called as error signal) as a means of control. Closed loop control system uses a measurement of the putput and feedback of this signal to compare it with the reference input. That means the output has an effect on the control action. Fig 1.3 shows the block diagram of the system.



*Fig. 1.3 Block diagram of closed loop control system.*

In this type of system the output is compared with the reference input, the output of the comparator is actuating or error signal which is input to the controller and the output of the controller is input to the process from which the desired output is obtained.

Examples of closed loop control systems are;

1. Room temperature control system,
2. Radar systems
3. Satellite systems.

### **1.3.1 Advantages of closed loop control systems**

1. The system output is insensitive to internal disturbances and variations in system parameter.
2. Optimization is possible
3. They are accurate
4. They are reliable because they adjust non linearities present in its element.

### **1.3.2 Disadvantages of closed loop control system**

1. They are complex in construction
2. They are more expensive.
3. They can become unstable under certain conditions.
4. More maintenance is required .

### **1.3.3 Components of closed loop control system**

1. **Error Detector:** Error detector is a device which is used to compare the feedback signal obtained from the output with the reference input signal and gives a deviation which is called 'actuating signal'.

2. **Controller:** The controller is a device which is used to produce the control signal as the input to the process or controlled system. So in other words we can say that controller is used for signal conditioning of actuating signal because actuating signal is of low power level so there is need to manipulate the signal by the controller.

3. **Process or controlled system:** The controlled system is the device whose condition is to be maintained at all times as desired and this corresponds to the desired output.

4. **Feedback element:** The feedback element is a device which is used to convert the output variable into some other variable which is compared with the reference input.

## 2.0 Speed Control

Speed control means intentional change of the drive speed to a value required for performing the specific work processes. This concept of speed control or adjustment should not be taken to include the natural change in speed, which occurs due to change in the load on the drive shafts. The desired change in speed is accomplished by acting accordingly on the drive motor or on the transmission connecting it to the unit it serves to drive. This may be done manually by the operator or by means of some automatic control device.

### 2.1 Need to speed control in machines

It is necessary to control the speed of machines because of the following reasons;

1. Some industrial devices may required continuous variation of speed for the whole range of work from zero to full speed.
2. Some other machines may require two or more fixed speeds,

3. Some machines may require creeping speed for adjusting or setting up the work.

## 2.2 Basic principle of speed control on no-load

The speed of a separately excited motor can be found from

$$\omega = \frac{V_a - R_a I_a}{K_v I_f} = \frac{V_a - R_a I_a}{K_v I_f / R_f} \text{-----} (2.1)$$

Where  $\omega$  = motor angular speed, or rotor angular frequency, rad/s

$K_v$  = Voltage constant, V/A – rad/s

$R_a$  = Armature resistance,  $\Omega$

$I_a$  = Armature current, A

$I_f$  Field current, A

We can notice from the equation (2.1) that the motor speed can be varied by;

1. Controlling the field current,  $I_f$ , known as field control; for fixed armature voltage;
2. Torque demand, which corresponds to an armature current  $I_a$ , for fixed field current known as 'Armature Control' or '<sup>Rheostatic</sup> Rheastatic control'; or
3. Controlling the armature voltage  $V_a$ , known as 'Voltage Control'.

The speed, which corresponds to the rated armature voltage, rated field current and rated armature current is known as the 'rated' (or base) or 'nominal' speed.

Other essential equations for separately excited motor are;

$$\text{Field voltage } V_f = R_f I_f \text{ - - - - -} (2.2)$$

$$\text{Back emf } e_g = K_v \omega I_f \text{ - - - - -} (2.3)$$

$$\text{Armature voltage } V_a = R_a I_a + E_g \quad (2.4)$$

$$\text{Torque developed } T_d = K_f I_f \quad (2.5)$$

$$\text{Power developed } P_d = T_d \omega \quad (2.6)$$

The speed of series motor can be determined from

$$\omega = \frac{V_a - (R_a + R_f) I_a}{K_b I_f} \text{ (rad/s)} \quad (2.7)$$

$$\text{But } 1 \text{ rad/s} = 9.55 \text{ rpm}$$

$$E_g = K \phi \omega I_a \quad (2.8)$$

$$V_a = (R_a + R_f) I_a + E_g \quad (2.9)$$

$$T_d = K_f I_f \quad (2.10)$$

The speed of series motor can be varied by controlling the;

1. Armature voltage  $V_a$  known as 'voltage control'.
2. Armature current which is a measure of the torque demand known as 'Armature control'.

### 2.2.1 Field control

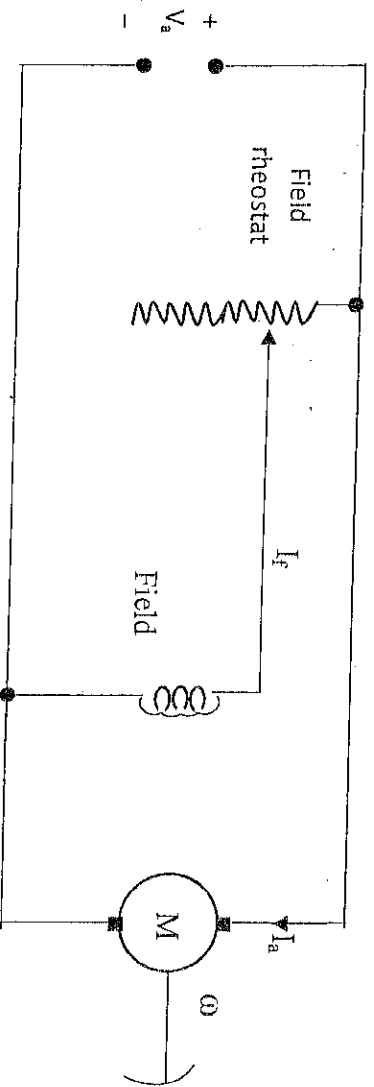


Fig. 2.2.1 Field control of D.C separated excited motor

The block diagram of field control of a.d.c separated excited motor is shown in Fig. 2.2.1. The supply voltage  $V_a$  which is the armature voltage is held constant while the field current  $I_f$  is varied by means of the field rheostat. As  $I_f$  varies, the speed of the motor is varied accordingly. From equation (2.1) given as;

$$\frac{V_a - R_a I_a}{K_v I_f / R_f}$$

If  $I_f$  increases the speed reduces. Alternatively, if  $I_f$  decreases the speed also decreases accordingly, hence providing a means of changing the speed.

### Examples:

A 622.25V shunt motor has an armature resistance of  $0.25\Omega$  and field resistance of  $175\Omega$  and the motor voltage constant is  $K_v = 1.4\text{V/A-rad/s}$ . The armature current corresponding to the load demand is  $I_a = 45\text{A}$ . The same 622.25V supplies the field, determine;

- The torque developed by the motor.
- Total input power from supply.

### Solution

$$V_A = V_f = 622.25\text{V}, \quad R_a = 0.25\Omega, \quad I_f = 175\Omega$$

$$I_a = 45\text{A} \text{ and } K_v = 1.4/\text{A-rads/s}.$$

$$\text{Field current } I_f = \frac{V_f}{R_f} = \frac{622.25}{175}$$

$$= 3.56\text{A}$$

$$\begin{aligned} \text{a) The torque developed } T_d &= K_v I_f I_a \\ &= 1.4 \times 3.56 \times 45 \\ &= 224.28 \text{N}_a\text{m} \end{aligned}$$

$$\text{b) Speed of the motor } \omega \text{ (rad/s) is}$$



$$\omega = \frac{V_a - R_a I_a}{K_b I_f} = \frac{622.25 - (0.25 \times 45)}{1.4 \times 3.56}$$

$$= \frac{622.25 - 11.25}{4.984} = \frac{611}{4.984}$$

$$= 122.69 \text{ rad/s}$$

$$= 122.69 \times 9.55$$

$$= 1171.69 \text{ rpm}$$

c) Assuming lossless supply system, the total power from the supply  $P_i$  is

$$P_i = V_a I_a + V_f I_f$$

$$= (622.25 \times 45) + (622.25 \times 3.56)$$

$$= 28801.25 + 2215.21$$

$$= 31066.46 \text{ W}$$

$$= 31 \text{ kW}$$

## 2.2.2 Armature control

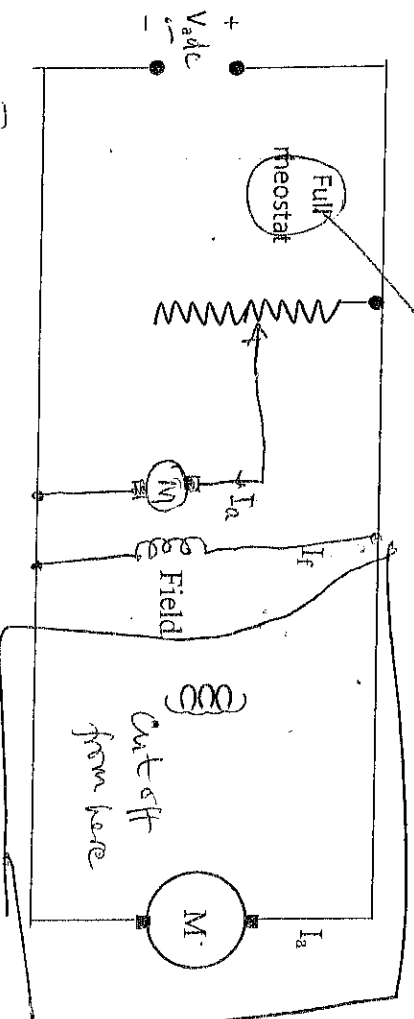
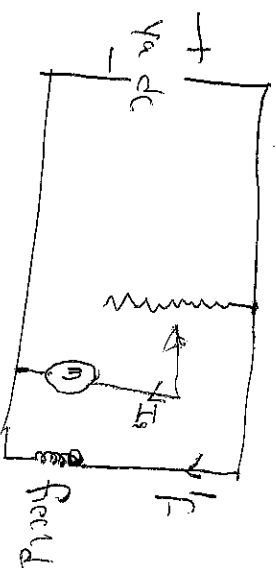


Fig. 2.2.2 : Armature control of D.C. separately excited motor

Fig. 2.2.2 Shows the working principle of armature control of d.c. separately excited motor. The supply voltage  $V_d.c$  which is the field Voltage is held constant while the armature current  $I_a$  is varied by the adjustment of the control resistor in the armature circuit. If the resistance introduced into the armature circuit by the control resistor is

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small, the armature current increases. Conversely, when the resistance introduced into the armature circuit is large, the armature current decreases. From equation (2.1) for calculation of the speed  $\omega$ , of the motor which is;

$$\omega = \frac{V_a - R_a I_a}{K_v I_f}, \text{ it means that as}$$

Armature current  $I_a$  increases the speed is reduced on the other hand, if the armature current is decreased, the speed increases accordingly, by this manipulation process the speed of the motor is controlled as desired.

### 2.2.3 Practical speed control system

The turntable is one practical application of speed control system. The turntable rotates a disk at a constant speed. For example, a CD player, a computer disc drive, and a phonograph record player all require a constant speed of rotation in spite of motor wear and variation and other components changes. The closed-loop control of the speed of a turntable is shown in Fig. 2.2.3.

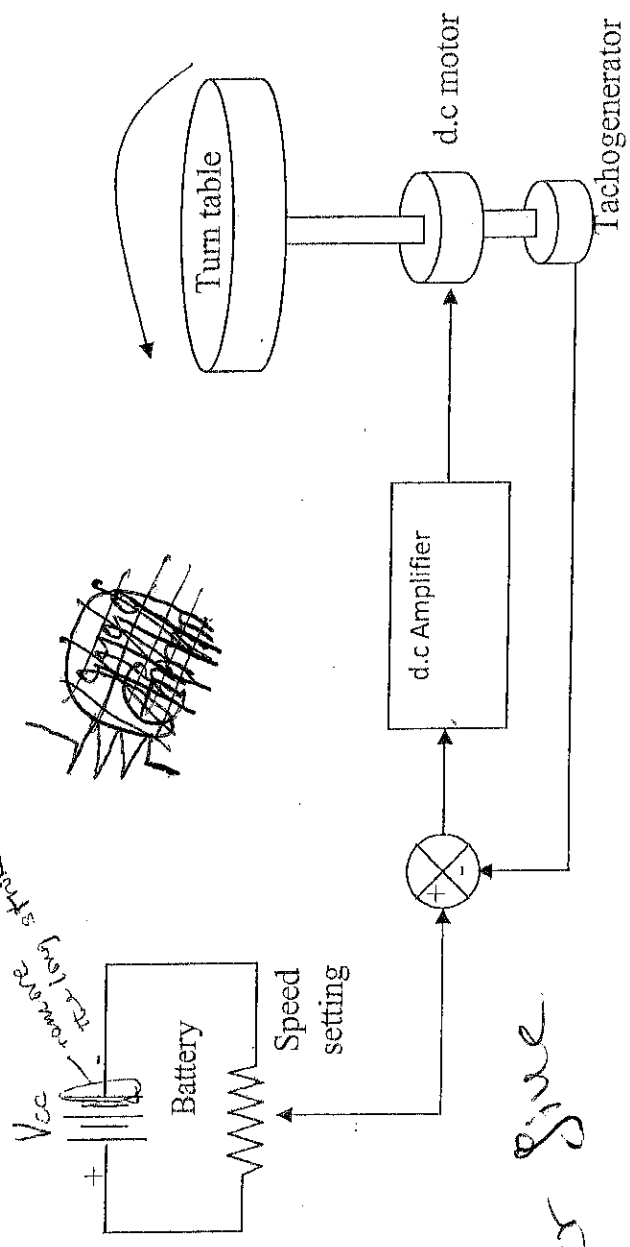
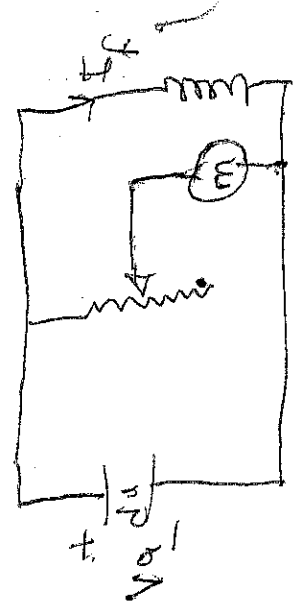


Fig. 2.2.3 Closed loop control of the speed of turntable

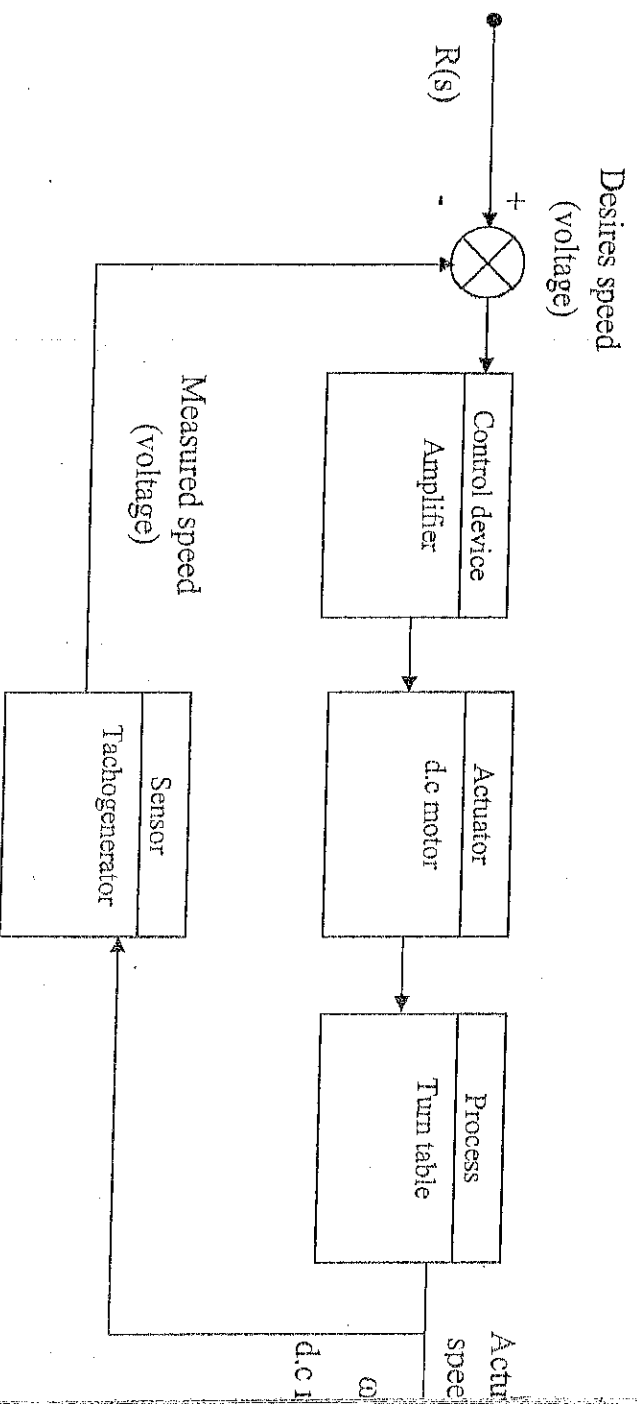


paire god  
Bless me  
as I fight my  
again,

Never Give  
up! still  
HORE  
dot.

The potentiometer connected across the battery is used to set a voltage that is proportional to the desired speed. The amplifier that feeds the motor provides the required power to drive the motor. The motor is the actuator that rotates the turntable at a speed proportional to the set reference voltage. The tachogenerator is a sensor that converts speed of the motor to an output voltage proportional to the speed of its shaft. The feedback voltage from the tachogenerator is fed into the comparator and used to compare with the set reference input. In the event of any deviation, an error signal is generated and applied to the amplifier which feeds the corrective signal to the motor to maintain its speed constant.

### 2.2.3 block diagram of turntable closed-loop speed control system



*Fig. 2.2.4 block diagram of turntable closed-loop speed control system*

**Mathematical model/transfer function of a control system**

Mathematical model is the mathematical representation of a process, device or concept by means of a number of variables which are defined to represent the inputs,

outputs and internal states of the device or process by a set of equation(s) describing the interactions of these variables.

**Transfer function** is defined as the ratio of the laplace transform of the output quantity to the laplace transform of the input quantity, with all initial conditions assumed to be zero. That is the transfer function of the system represents the relationship between the input and output variables which describes the dynamics of the system under consideration.

### 2.2.5 Transfer function of field controlled motor

The wiring diagram of a d.c motor is shown in fig. 2.2.6 below.

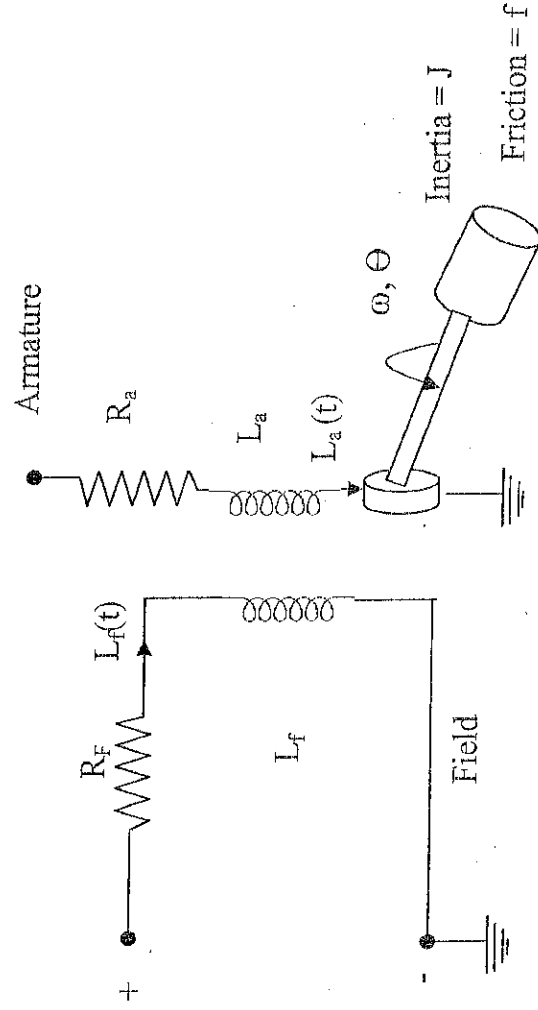


Fig. 2.2.6 (Wiring diagram of a d.c motor)

$e(t)$  = Field voltage  
 $T_m$  = Motor torque  
 $K_m$  = Motor constant

|    |          |   |                                                                        |
|----|----------|---|------------------------------------------------------------------------|
| ng | $T_d$    | = | Disturbance torque                                                     |
| ut | $\theta$ | = | angular displacement                                                   |
| ns | $f$      | = | velocity function                                                      |
| he | $T_d$    | = | disturbance torque                                                     |
| of | $\omega$ | = | angular speed                                                          |
|    | $E(s)$   | = | $(R_f + L_f) I_f(s)$ and $I_f(s) = \frac{E(s)}{(R_f + L_f s)}$ (2.2.7) |
|    | $R_f$    | = | Field resistance                                                       |
|    | $L_f$    | = | Field inductance                                                       |
|    | $I_f$    | = | Field current                                                          |
|    | $J$      | = | Moment of inertia of motor                                             |
|    | $I_a$    | = | armature current                                                       |

The disturbance torque is often negligible but must be considered in systems subjected to external forces such as antenna that experiences wind-gust forces.

For this consideration,  $T_d$  is neglected, hence, motor torque  $T_m$  must be equal to load torque  $T_L$ . Hence,

$$\frac{Jd\omega}{dt} + f\omega = T_m = K_T i_f \quad \text{Laplace transform} \quad (2.2.8)$$

This laplacetransform of equation (2.2.8) is given by

$$Js\omega(s) + f\omega(s) = T_m(s) = K_T I_f(s)$$

$$\text{Or } \omega(s) [J_s + f] = T_m(s) = K_T I_f(s) \quad (2.2.9)$$

$$\omega(s) = \frac{K_T I_f(s)}{(J_s + f)} \quad (2.2.10)$$

Equation (2.2.10) is the output variable.

Considering field control, the input variable is the field voltage  $E_s(s)$  given by equation (2.2.7). the transfer function of the system is

$$\frac{\omega(s)}{E(s)} = \frac{K_T I_f(s)}{(J_s + f)} = \frac{K_T I_f(s)}{(J_s + f)} \times \frac{1}{(R_f + L_f s) I_f(s)}$$

$$(R_f + L_f s) I_f(s)$$

$$\text{Hence } \frac{\omega(s)}{E(s)} = \frac{K_T}{(J_s + f)(L_f s + R_f)} \quad (2.2.11)$$

Hence,

$$G(s) = \frac{\omega(s)}{E(s)} = \frac{K_T / f R_f}{\left[\left(\frac{J}{f}\right) + 1\right] \left[\left(\frac{L_f}{R_f}\right) s + 1\right]} \quad (2.2.12)$$

$$\text{Let } K_m = \frac{K_T}{f R_f} \quad T_{me} = \frac{J}{f} \text{ and}$$

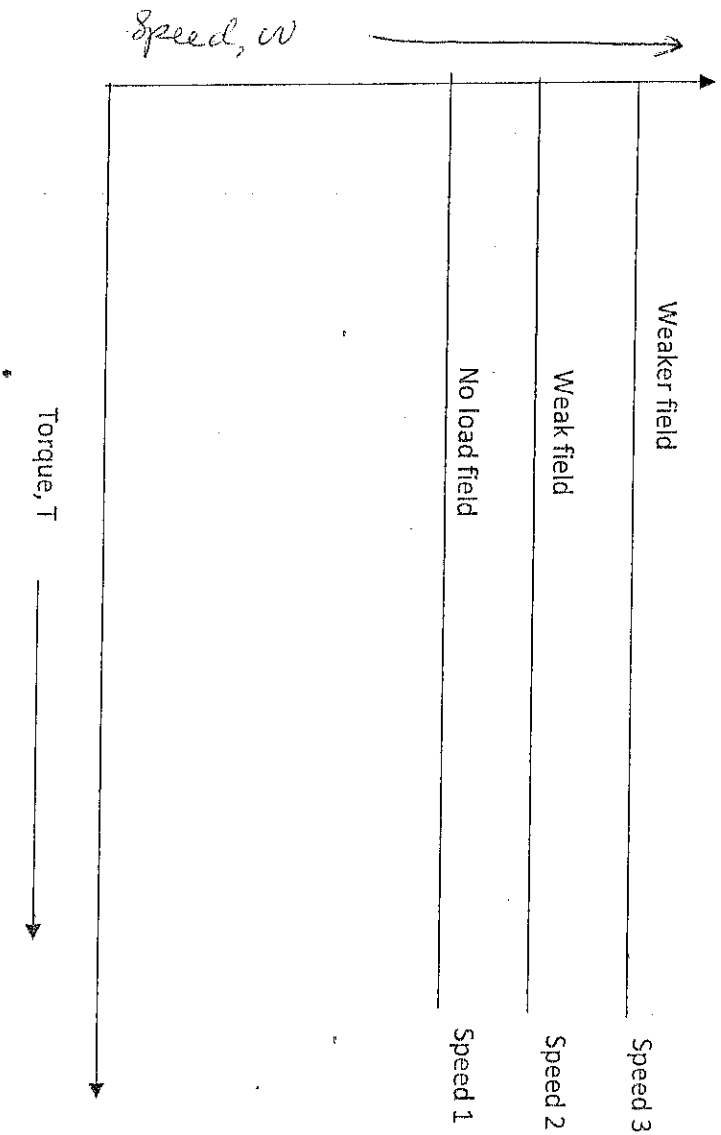
$$Y_f = \frac{L_f}{R_f}$$

## 2.6 Merits and demerits of field control over armature control

| Merits                                                                                                                                                                                                                                | Demerits                                                                                                                                                                                                                                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ol style="list-style-type: none"> <li>1. Field control is economical</li> <li>2. Field control is more efficient</li> <li>3. It gives speeds above the nominal speed.</li> <li>4. Field control is more convenient to use</li> </ol> | <ol style="list-style-type: none"> <li>1. Field control cannot give speed below the nominal speed.</li> <li>2. Field control gives unsatisfactory commutation because armature reaction is greatly manifested during weaker fields</li> </ol> |

## by Effects of Load Torque on Speed

### Field Control



Then substituting  $K_m$ ,  $Y_{me}$  and  $T_f$  into equation (2.2.12) gives,

$$G(s) \frac{\omega(s)}{E(s)} = \frac{K_m}{(Y_{me}s+1)(Y_{fs}+1)} \quad (2.2.13)$$

Where;

$K_m$  = motor given constant

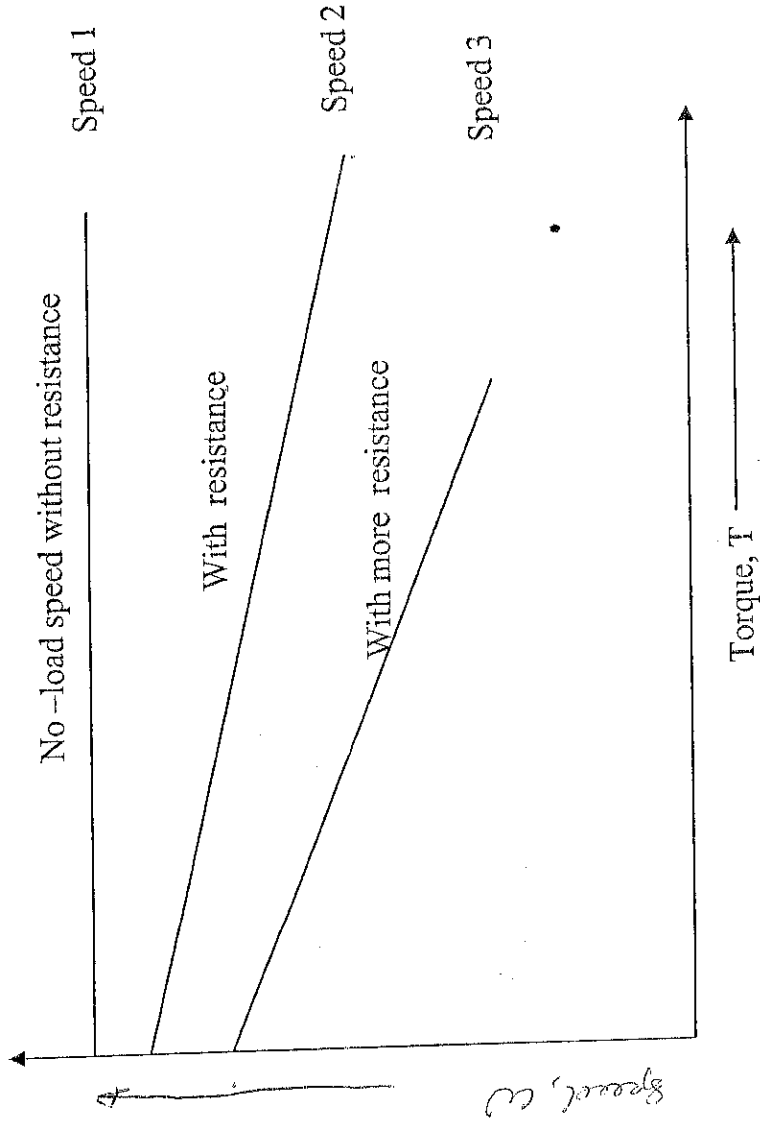
$Y_f$  = time constant of the field current.

$Y_{me}$  = mechanical time constant.

FIG. 2.7.1 shows the graph of speed-versus-torque with a constant armature voltage. From the graph, the following conclusion can be drawn;

1. Speed increase as the field gets weaker and remains fairly constant at the set speed despite change in the load, torque.
2. Speed increase above the no-load (nominal) speed is attainable while speed variation below the no-load speed is not possible.

### 2.7.2 Armature control



*Fig. 2.7.2. Effect of Load Torque on speed*

Fig. 2.7.2 is a graph of speed-versus-torque with a constant field constant. From the graph the following inferences can be drawn;

1. At the no-load speed that is without resistance introduced on the armature circuit the speed is highest and constant.
2. As resistance is introduced in the armature circuit, the speed drops, which means a change in speed can be achieved, in this way, hence speed 2 and 3 shown.



ne set  
speed

3. With increase in the load torque the speed drops further from the set value
4. Speed variations whom the no-load speed are achievable while those above the no-load speed are impossible.

## 2.8 Velodyne speed control system

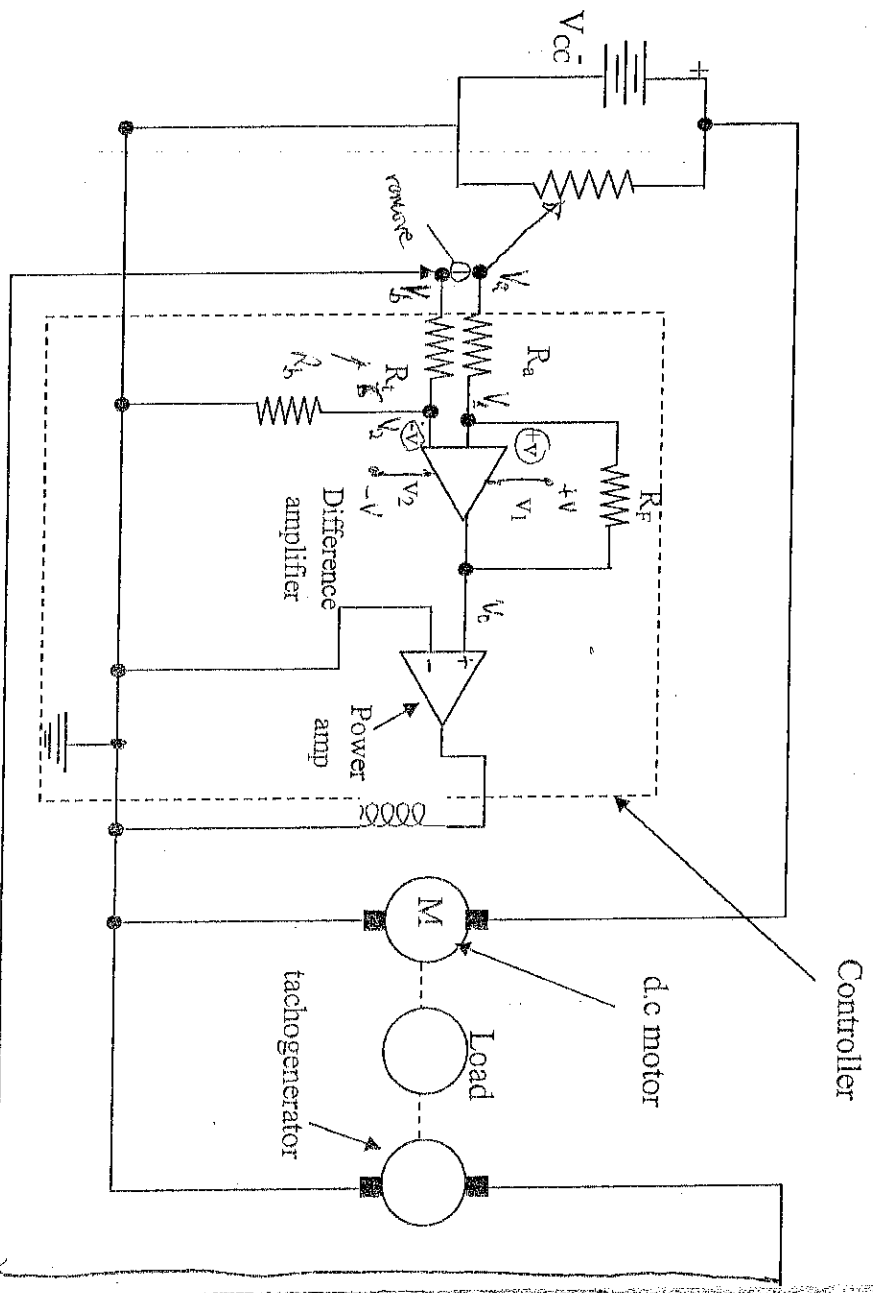


Fig. 2.8 Velodyne speed control system

Fig. 2.8 is a schematic diagram of the principle of operation of the velodyne speed control system. It consists of a d.c motor whose field is controlled electronic amplifier. In the arrangement, the motor armature voltage is maintained constant and its output shaft is mechanically coupled to a tachogenerator. A reference signal  $V_a$  is derived from a potential divider connected across a constant d.c source  $V_{cc}$ . the controller receives the error signal  $(V_a - V_b)$ . The aim of the system is to maintain the

speed constant at a level proportional to  $V_a$  regardless of load torque. Deviation in speed is connected to its equivalent by the techogenators and feedback to the controller as  $V_b$ . the difference volage ( $V_s - V_b$ ) is amplified ad fed to the field for correction of the speed. This speed control system finds application in instrument series. If  $R_a = R_b$  and  $R_f = R_g$ , them,

$$V_0 = \frac{R_f}{R_a} [V_a - V_b]$$

### 2.8.1 Merits and demerits of veladyne control

| Merits                                                                                                                                                                                                                                                                                          | Demerits                                                                                                                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ol style="list-style-type: none"> <li>1. Very small field currents are used for control, hence, it does not waste power and is efficient.</li> <li>2. It is less expensive</li> <li>3. It is very portable</li> <li>4. It provides immediate correction for unexpected disturbances</li> </ol> | <ol style="list-style-type: none"> <li>1. It cannot be used for heavy loads</li> <li>2. Change in speed is not rapid</li> <li>3. Design and construction are complex</li> </ol> |

### 2.9 Ward-Leonard speed control system

The schematic diagram of Ward-Leonard speed control system is shown in Fig. 2.9  $M_1$  is the main motor whose speed control is required. The field of the motor is permanently connected across the d.c supply lines. By applying a variable voltage across its armature, any desired speed can be obtained. The variable voltage is supplied by a motor-generator set which consists of either a d.c or an a.c motor  $M_2$  directly coupled to generator G.

The motor  $M_2$  runs at an approximately constant speed. The output voltage of  $G$  is directly fed to the main motor  $M_1$ . The voltage of the generator can be varied from zero up to its maximum values by means of its field regulator. By reversing the direction of the field current of  $G$  by means of reversing switch  $RS$ , generator voltage can be reversed and hence the direction of rotation of  $M_1$ . It should be noted that motor-generator set always runs in the same direction.

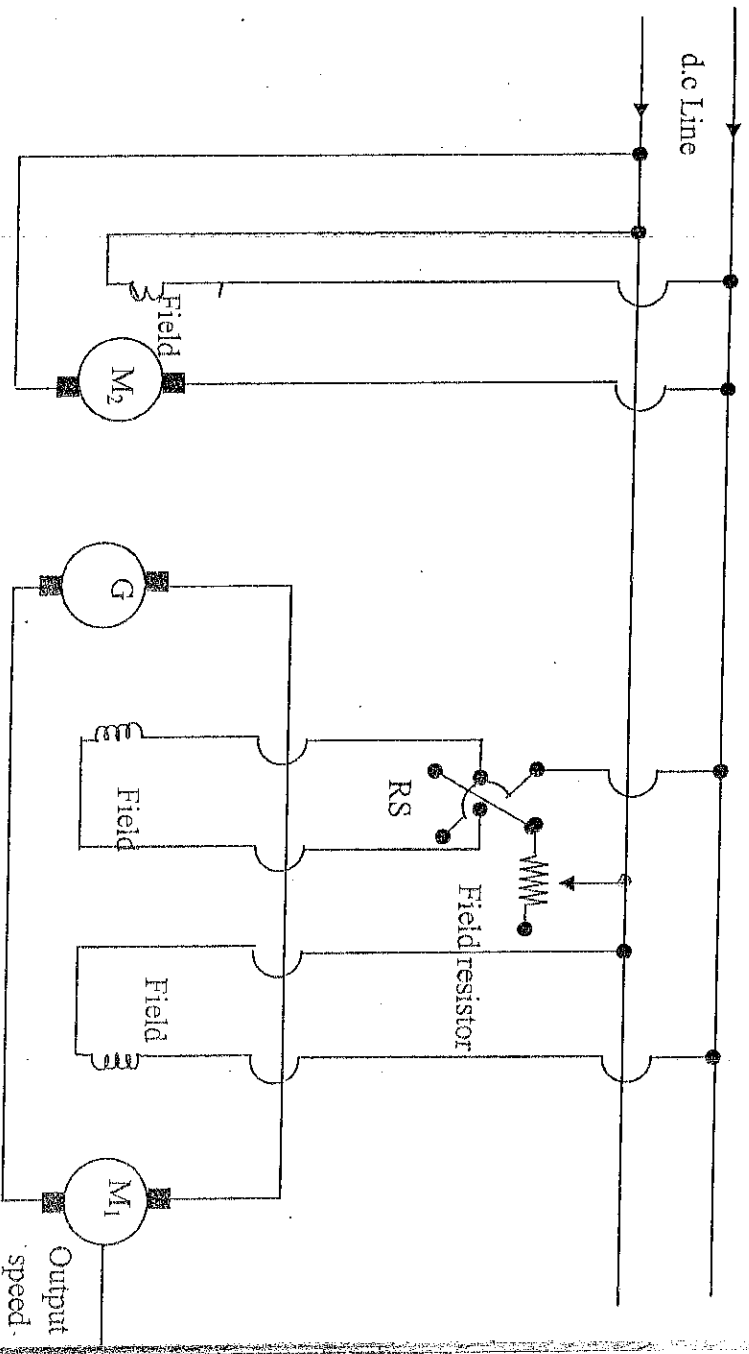


Fig. 2.9. Ward-Leonard speed control system

### 2.9.1 Merits and demerits of Ward-Leonard speed control

| Merits                                                                                                        | Demerits                                                           |
|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| 1. Very fine speed control over the whole range from zero to nominal speed in both directions can be obtained | 1. Two extra machines are required, so arrangement is costly.      |
| 2. Uniform acceleration can be obtained                                                                       | 2. Low overall efficiency of the system, especially at high loads. |
| 3. Speed regulation is good                                                                                   |                                                                    |

### 2.9.2 Applications of Ward-Leonard Speed Control

The system of speed control is best suited where almost unlimited speed control in either direction of rotation is required such as;

1. Electric excavator
2. Cranes
3. Mine hoists
4. Diesel electric <sup>propulsion</sup> propulsion of ships
5. Main drives in steel mills.
6. Main drives in blooming and paper mills.

### 2.10 Thyristor Speed Control System

The average output voltage of a thyristor-controlled rectifier can be changed by changing its delay angle,  $\alpha$  and hence the armature voltage of the d.c motor can be adjusted to control its speed. When run on a d.c supply, the armature d.c voltage can be changed with the help of a transistor chopper circuit which can be made to interrupt d.c supply at different rates to give different values of the d.c voltage.

## Thyristor full-wave speed control

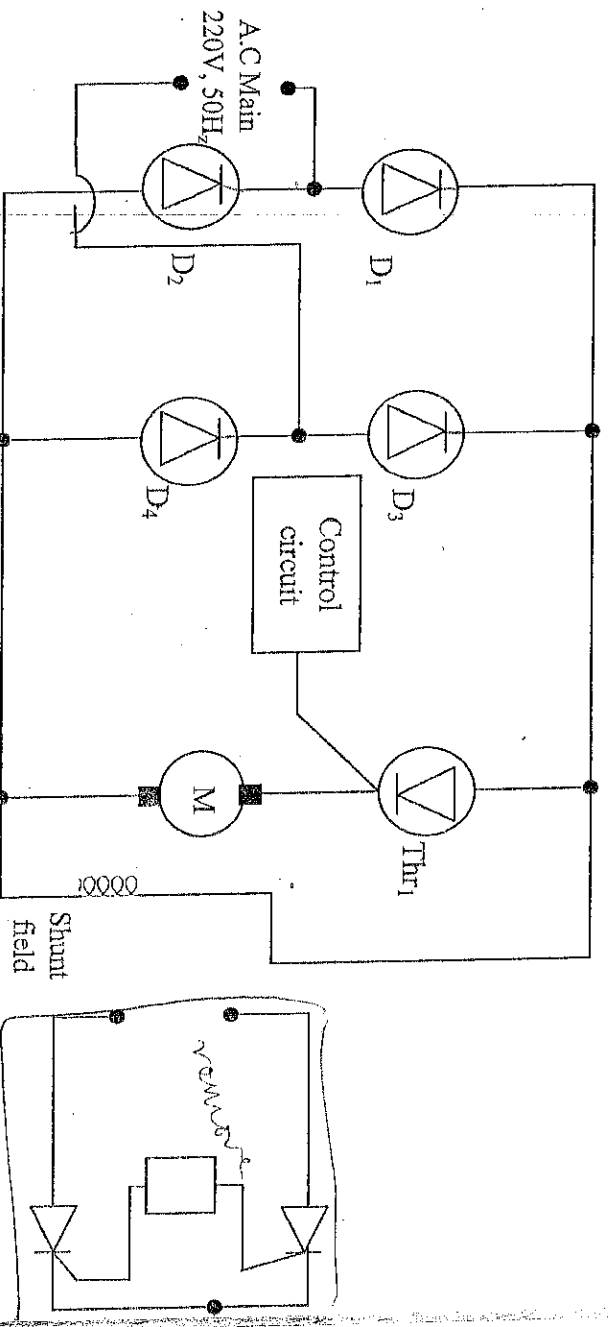


Fig. 2.10(a) Thyristor full-wave speed control

Thyristor full-wave speed control system is shown in fig. 2.10(a). The full-wave rectifier are  $D_1$ ,  $D_2$ ,  $D_3$  and  $D_4$ . The output of the rectifier is fed to the field winding of the motor, and this field is maintained constant. The rectifier output voltage is also applied to the armature of the motor through the thyristor  $Thr_1$ . The thyristor is turned ON at a particular delay angle  $\alpha$  and allowed to conduct for a period corresponding to a conduction angle set by the control circuit d. After the delay angle  $\alpha$ , voltage is applied to the armature of the motor which determines the speed. If the delay angle is small, more voltage is applied to the motor thereby increasing the speed. Conversely, if the delay angle is large, less voltage is applied to the

armature thereby reducing the motor speed. With this method, the speed of the motor is controlled. The waveform across the motor is shown in fig. 2.10.1(b)

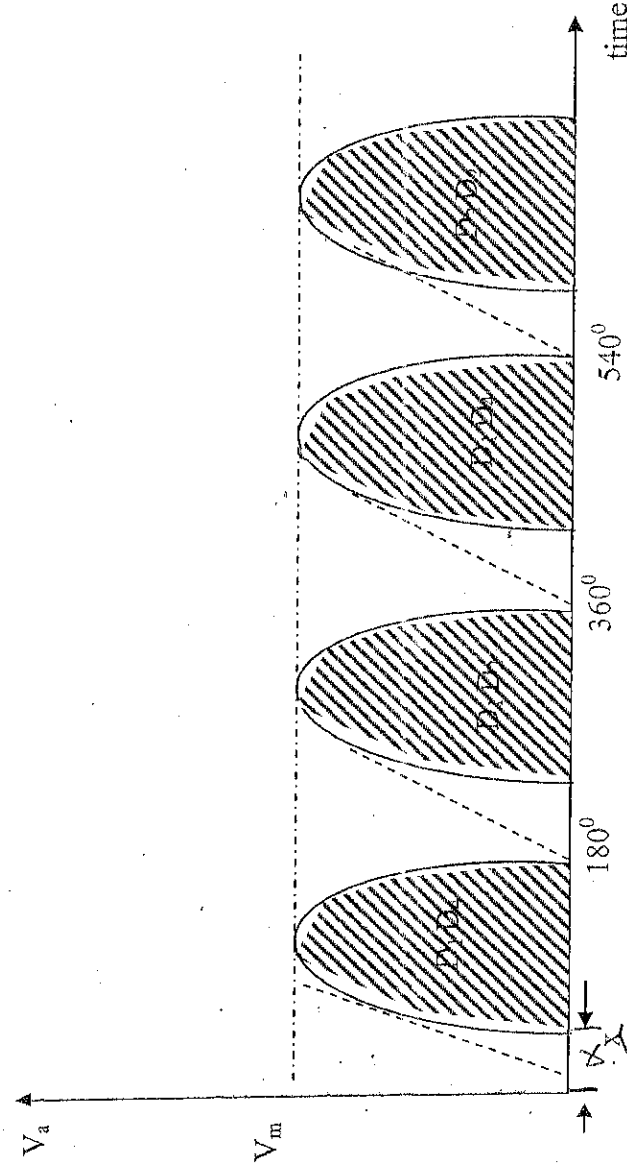


Fig. 2.10.1(b) Load voltage waveform

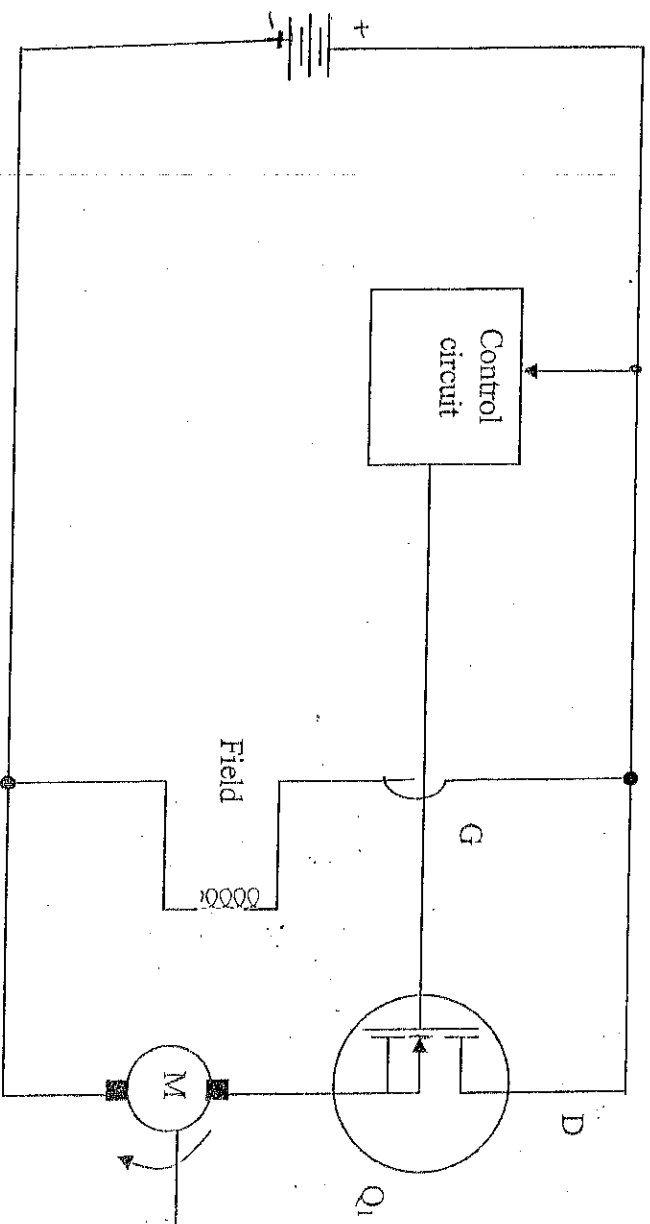
$$V_a = \frac{V_m}{\pi} (1 + \cos \alpha) \text{ and } I_a = \frac{V_m}{\pi R L_a} (1 + \cos \alpha) \quad (2.10.1)$$

Where:  $V_a$  = average d.c voltage of armature

$I_a$  = average d.c current of armature.

$V_m$  =  $(\frac{\sqrt{2}}{2}) \times \text{a.c supply voltage}$

$\alpha$  = the delay angle or firing angle



*Fig. 3.10.2 Chopper speed control*

Fig. 2.10.2(a) shows an arrangement for transistor chopper speed control system. The supply d.c voltage is applied permanently at the drain of the FET. The supply voltage to the motor is controlled by the FET,  $Q_1$ . The control circuit switches the FET ON and OFF very rapidly. This action interrupts the d.c supply at a regular rate (or frequency) in order to produce a lower (means) d.c voltage supply to the motor armature. The average d.c voltage applied to the armature of the motor determines its speed.

The mean value of the output voltage produced by chopping action is shown in fig. 2.10.2(b)

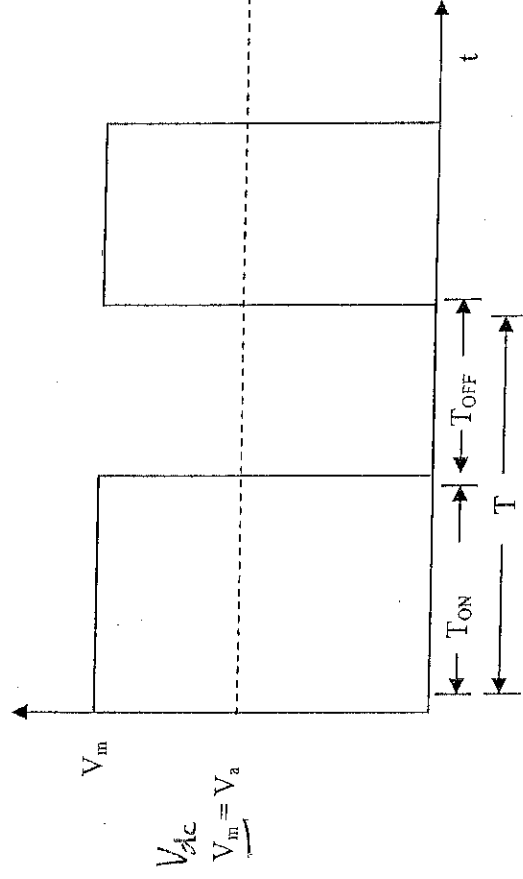


Fig. 2.10(b) Thyristor  
chopper d.c output  
waveform

The equation for the mean value of the output voltage is given by;

$$V_{dc} = V_a = V_m \left[ \frac{T_{ON}}{T_{ON} + T_{OFF}} \right] = V_m \left[ \frac{T_{ON}}{T} \right] \quad (2.10.2)$$

$\frac{T_{ON}}{T}$  = the duty cycle

The speed equation for thyristor controlled speed system is given by;

$$\omega \text{ (rad/s)} = \frac{V_a - R_a I_a}{K_v I_f} = \frac{E_b}{K_v I_f} \quad (2.10.3)$$

Where:

$V_a$  = armature voltage (Volts)

$R_a$  = armature resistance ( $\Omega$ )

$I_a$  = armature current (A)

$K_v$  = voltage constant  $\frac{V}{A}$  - rad/s

$I_f$  = field current  $\frac{A}{A}$

$\frac{\omega}{C_v}$  = speed in rad/s

$E_b$  = back E.m.f



### Examples:

A thyristor full-wave speed controlled d.c shunt motor is supplied with 220v, 50Hz a.c. If the <sup>the delay</sup> angle  $\alpha$  is  $10^\circ$ ,  $R_a = 0.25\Omega$ ,  $I_a = 42A$ , voltage constant  $K_v = 0.7032V/A\text{-rad/s}$ ,  $V_f = 198v$ ,  $R_f = 147\Omega$ . Calculate;

- The average d.c voltage to the motor
- The speed of the motor in rpm.

### Solution:

Given; supply voltage  $V_s = 220v$ , a.c

$$A = 10^\circ$$

$$R_a = 0.25\Omega$$

$$I_a = 42A$$

$$K_v = 0.7032V/A\text{-rad/s}$$

$$V_f = 198V$$

$$R_f = 147\Omega$$

- a. The average d.c voltage to the motor  $V_a$  is

$$V_m = \sqrt{2} \hat{V}_s = \sqrt{2} \times 220 = 311.13$$

$$\begin{aligned} V_a &= \frac{V_m}{\pi} (1 + \cos \alpha) = \frac{311.13}{\pi} (1 + \cos 10^\circ) \\ &= 99.03 (1 + 0.9848) \\ &= 196.55v \end{aligned}$$

- b. The speed of the motor;

$$I_f = \frac{V_f}{R_f} = \frac{198}{147} = 1.35A$$

$$E_b = V_a - R_a I_a = 196.55 - 0.25 \times 42$$

$$= 196.55 - 10.5$$

$$= 186.05V$$

$$\omega = \frac{E_b}{K_v I_f} = \frac{186.05}{0.7032 \times 1.35} = \frac{186.05}{0.9493}$$

$$= 196 \text{ rad/s}$$

$$\text{But } \omega_{\text{rpm}} = \frac{\omega}{\omega} \times 9.55$$

$$= 196 \times 9.55$$

$$= 1871.7 \text{ rpm}$$

### 2.10.3 Exercises

1. A thyristor full-wave controlled d.c shunt motor is supplied with 220V, 50Hz

ac. If the delay angle  $\alpha$  is  $13^\circ$ ,  $R_a = 0.27\Omega$ ,  $I_a = 50A$ , voltage constant,  $k_v = 0.3145V/A \text{ rad/s}$ ,  $V_f = 200\Omega$ . Calculate;

- The average d.c voltage to the motor
- The speed of the motor in rpm

2. A thyristor chopper speed controlled d.c shunt motor is supplied with 300V

d.c. If the chopper has  $T_{ON} = 15 \text{ mill.-second}$  and  $T_{OFF} = 5 \text{ milli-second}$ ,  $R_a = 0.25\Omega$ ,  $I_a = 45A$ , voltage constant  $k_v = 0.413V/A \text{ rad/s}$ ,  $V_f = 250V$ ,  $R_f = 250\Omega$ . Calculate;

- The average d.c voltage to the motor
- The speed of the motor in rpm.

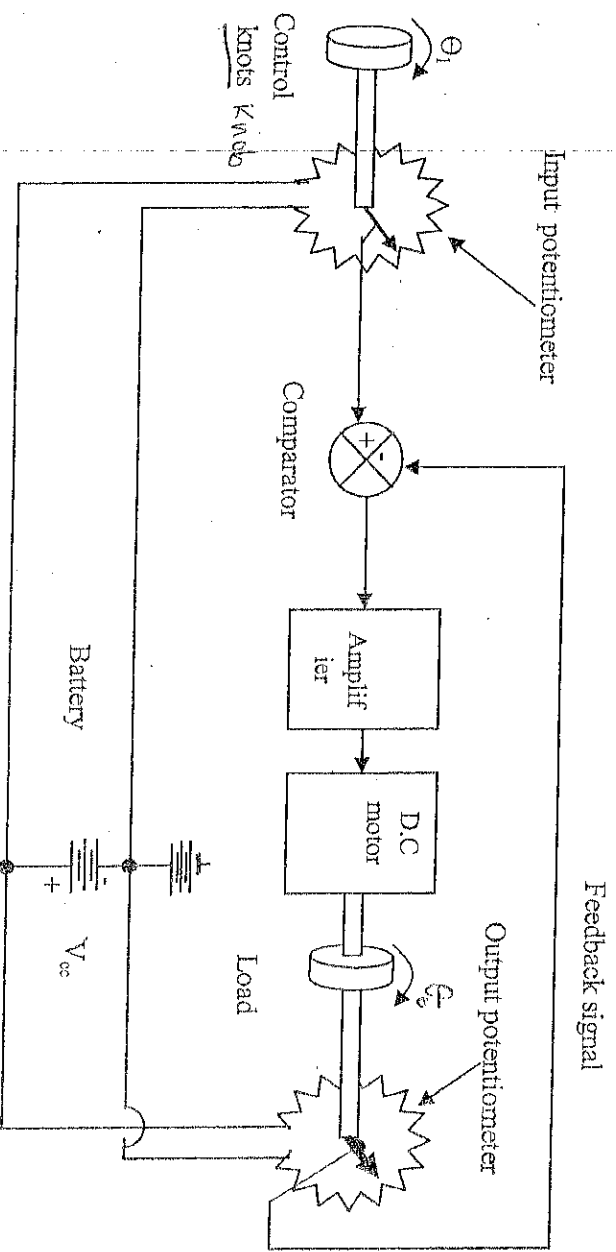
### 3.0 Need for position control system

Need for position control arises because,

- Numerical control machine tools require precise positioning of a work piece.
- Robotic arms move in specified paths that require sensing and control of positions.

3. Some sequential controllers must sense the presence of a part and determine its exact location.
4. There is problem of mechanical position or translation.
5. Position control of heavy masses occur in machine tool control, automatic steering of aircraft, ships, guided weapons, and space craft.
6. Automatic positioning of radar aerials, guns and missile launching and guidance system, and radio telescopes regularly arise in the industry.

### Position control system



*Fig. 3.1 Position control system*

Fig. 3.1 shows the diagram for a position control system, two potentiometers, namely, the input and output potentiometers are connected in parallel across the same power source  $V_{cc}$ . The input position  $\theta_1$  is set by the control knob whose equivalent voltage is tapped from the input potentiometer and fed to the comparator through the '+' terminal. This voltage is amplified by the amplifier and fed to the

motor which turns the load to an output position  $\Theta_i$ . The shaft to the load is extended and keyed to the wiper of the output potentiometer. The output position  $\Theta_0$  is converted to its equivalent voltage by the potentiometer and fed back to the comparator input terminal. The feedback voltage which corresponds to output position  $\Theta_i$  compares with the set input (or reference input) and if there is a deviation an error signal is produced and used to drive the motor to a direction that will correspond to the desired input position. By this procedure, the output position is automatically maintained constant until another set position is input.

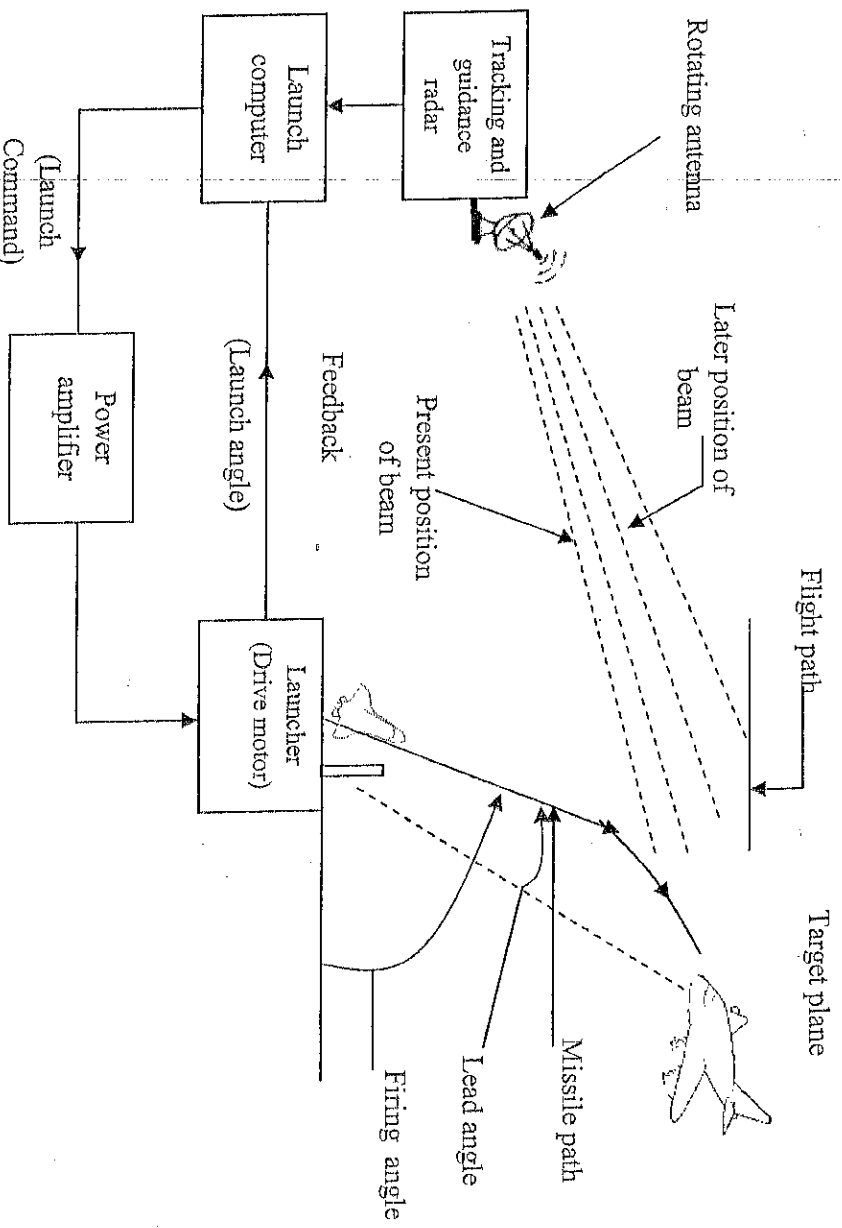
### 3.2 Practical position control systems

Two practical position control systems are missile launching and guidance system and satellite antenna position control system.

#### 3.2.1 Missile launching and guidance system

Fig. 3.2.1 is the block diagram of missile launching and guidance system. This is a sophisticated example of military application of position feedback control. The target plane is sited by a rotating radar antenna which then locks in and continuously tracks the target. Depending upon the position and velocity of the plane as given by the radar output data, the launch computer calculates the firing angle in ~~the~~ <sup>terms of a</sup> launch command signal, which when amplified through a power amplifier drives the launcher (drive motor). The launcher angular position is fed back to the launch computer and the missile is triggered as soon as the error between the launch command signal and the missile firing angle becomes zero. After being fired the missile enters the radar beam which is tracking the target. The control system contained within the missile now receives a guidance signal from the beam which automatically adjusts the control surfaces of the missile such that the missile rides along the beam, finally homing on the target.

ended  $\partial_0$  is the output radiation it will be ion is system:



**Fig. 3.2.1** *Missile launching and guidance system*

## 2.2 Satellite antenna position control system

System concept for controlling the position of satellite antenna is shown in Fig. 3.2.2. It consists two potentiometers connected in parallel across the same power source  $V_{cc}$ . The two potentiometers serve as input and output transducer respectively. The input position  $\Theta_i$  is set by adjustment of the control knob whose shaft is keyed to the wiper of the input potentiometer. As the knob is adjusted to the desired position  $\Theta_i$ , the wiper of the potentiometer moves to its equivalent voltage. This voltage is tapped and applied to “+” terminal of the comparator. The voltage is amplified by the amplifier and fed to the motor which turns the satellite antenna attached to its shaft to the desired output position  $\Theta_o$ . Through gear train linked to

the wiper of the output transducer, the wiper adjusts to the voltage corresponding to  $\Theta_0$  as the motor turns the antenna. The voltage produced in output potentiometer is <sup>signal</sup> tapped and fed back to the “-” terminal of the comparator. This feedback signed compares with the set input (or desired reference input) and if there is a deviation, an error signal is produced and used to drive the motor to a direction that will force the antenna to move to the desired output position  $\Theta_0$ , that corresponds to the setting. With this feedback position control system the satellite antenna can be positioned to track the required satellite.

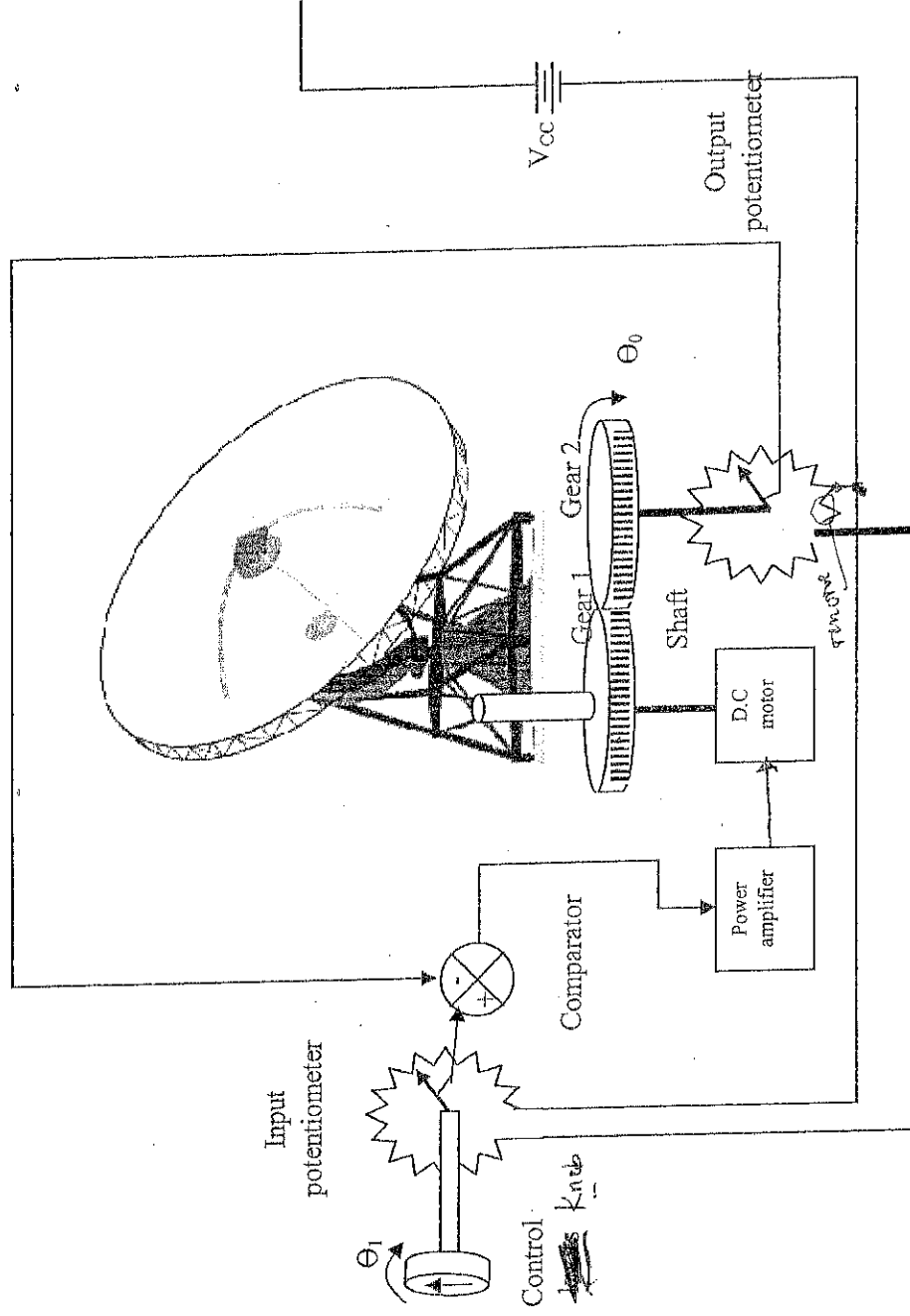


Fig. 3.2.2 satellite antenna position control system

### 3.3 Procedure in designing and analyzing a control system

The first step in analyzing a control system is to formulate the "physical model". An idealized physical system is called a physical model. From the physical model, the mathematical model is derived. Mathematical model of a system is the mathematical representation of the physical model through use of appropriate physical laws. The mathematical model that gives a greater insight into the dynamic behavior of physical system is selected. Mathematical models of most physical systems are characterized by differential equations. When the mathematical model of a physical system is solved for various input conditions, the result represents the dynamic response of the system.

#### 3.3.1 Physical models of d.c motor used for position and speed control

##### (a) Field control system

##### (i) Physical model of the system

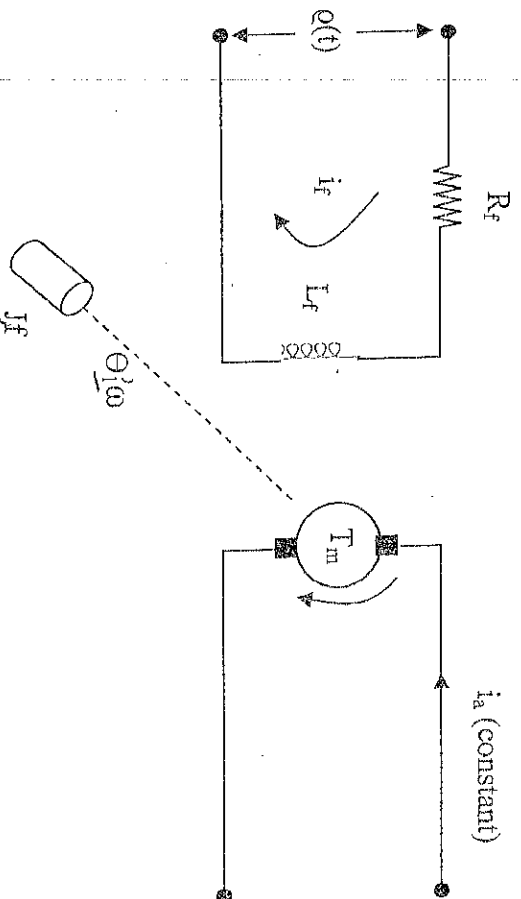


Fig. 3.3.1 (a) Field controlled d.c motor

In this system,

$R_f$  = Field winding resistance (ohms)

$L_f$  = Field winding inductance (henry's)

$e(t)$  = Field voltage (volts)

$i_f$  = Field current (amperes)

$T_m$  = Torque developed by motor (newton-m)

$J$  = equivalent moment of inertia of motor and load referred to motor shaft  
( $k_g\text{-m}^2$ )

$f$  = equivalent viscous friction coefficient of motor and load referred to motor shaft (newton-m/rad/s)

$\Theta$  = angular displacement of motor shaft (rad)

$\omega$  = angular speed of motor shaft (rad/s)

$\frac{L_a}{L_e} \frac{L_a}{L_e} \frac{L_a}{L_e} \frac{L_a}{L_e}$  = armature current (amperes) fed constant.

#### Mathematical model of the system

$$T_m = \frac{k K_f L_f L_a}{k K_f L_f L_a} = k_T i_f \quad (1)$$

Where  $k_T$  is a constant

The equation for the field circuit is

$$L_f \frac{di_f}{dt} + R_f i_f = e \quad (2)$$

The torque equations for position and speed control systems are given in equation (3) and (4) respectively



$$J \frac{d^2 \theta}{dt^2} + f \frac{d\theta}{dt} = T_m = k_T i_f \quad (3)$$

$$J \frac{d\omega}{dt} + f \omega = T_m = k_T i_f \quad (4)$$

### Position control consideration

The laplace transforms of eqns(2) and (3) are used as follows;

$$L_s I(s) + R_f I_f(s) = E(s)$$

$$\text{Or } (L_s + R_f) I(s) = E(s) \quad 5$$

$$J s^2 \Theta(s) + f s \Theta(s) = T_m(s) = k_T I_f(s)$$

$$\text{Or } (J s^2 + f s) \Theta(s) = T_m(s) = k_T I_f(s) \quad 6$$

From equation (6)

$$\Theta(s) = \frac{k_T I_f(s)}{(J s^2 + f s)} \quad 7$$

From the above equations, the transfer function of the system is obtained as

$$\frac{\Theta(s)}{E(s)} = \frac{\frac{k_T I_f(s)}{J s^2 + f s}}{(L_f s + R_f) I_f(s)}$$

$$= \frac{k_T}{(J s^2 + f s)(L_f s + R_f)}$$

Hence,

$$\frac{\Theta(s)}{E(s)} = \frac{k_T}{s(J s + f)(L_f s + R_f)} \quad 8$$

Dividing top and bottom of equation(s) by  $R_f$  gives

$$\frac{\Theta(s)}{E(s)} = \frac{K_T/R_f}{s(U_s + f)[s(\frac{L_f}{R_f}) + 1]} \quad 9$$

Again dividing the numerator and denominator of equation (9) by  $f$  gives;

$$\frac{\Theta(s)}{E(s)} = \frac{K_T/R_f f}{s \left[ s \left( \frac{L_f}{f} \right) + 1 \right] \left[ s \left( \frac{L_f}{R_f} \right) + 1 \right]} \quad 10$$

$$\text{Let } K_m = \frac{K_T}{R_f f} = \text{motor gain constant}$$

$$Y_f = \frac{L_f}{R_f} = \text{time constant of field circuit and}$$

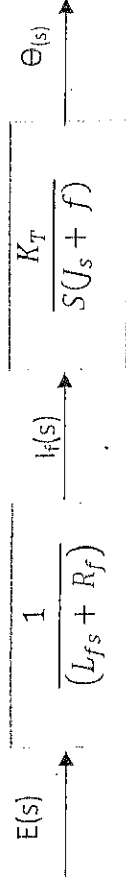
$$Y_m = \frac{J}{f} = \text{mechanical time constant, then equation(s) becomes,}$$

$$\frac{\Theta(s)}{E(s)} = \frac{K_m}{s(Y_m s + 1)(Y_f s + 1)} \quad 11$$

Equation (11) gives the Laplace transform of field control d.c motor used for position control, that's;

$$\frac{\Theta(s)}{E(s)} = \frac{K_m}{s(Y_m s + 1)(Y_f s + 1)} \quad (12)$$

Using equation(8) the block diagram of the system is shown below;



### Speed control consideration

The laplace transform of equation (2) and (4) are used as follows'

The laplace transform of equation (2) is equation (5) given as

$$(L_f s + R_f) I_f(s) = E(s)$$

Equation (4) is

$$J \frac{d\omega}{dt} + F\omega = T_m = K_T I_f \text{ whose laplace transform is;}$$

$$J_s \omega(s) + f\omega(s) = T_m(s) = K_T I_f(s)$$

$$\text{Or } (J_s + f) \omega(s) = T_m(s) = K_T I_f(s) \quad (13)$$

From equation (13) we obtained that

$$\omega(s) = \frac{K_T I_f(s)}{(J_s + f)} \quad (14)$$

From the foregoing, the transfer function of field controlled speed system is given as,

$$\frac{\omega(s)}{E(s)} = \frac{\frac{K_T I_f(s)}{(J_s + f)}}{(L_f s + R_f) I_f(s)}$$

used for

$$\frac{\omega(s)}{E(s)} = \frac{K_T}{(J_s + f)(L_f s + R_f)} \quad 15$$

$$\text{Let } K_m = \frac{K_T}{R_f f} = \text{motor gain constant,}$$

$$T_f = \frac{L_f}{R_f} = \text{Time constant of field circuit}$$

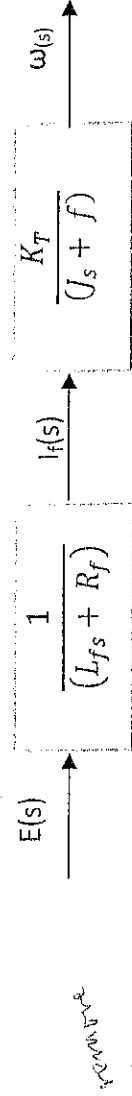
$$Y_f = \frac{I_f}{f} = \text{mechanical time constant, then equation (5) becomes;} \quad (15)$$

$$\frac{Y_f}{E(s)} = \frac{K_m}{(Y_m s + 1)(Y_f s + 1)} \quad 16$$

Equation (16) gives the transfer function of field controlled d.c motor used for speed control.

(15)

Using equation (16) the block diagram of the system is as shown below.



From the above equations, the transfer function of the system is obtained as;

The schematic diagram of a control transmitter (CT) and control receiver (CR) synchro-system is shown in fig. 3.3. The rotors are connected in parallel across the a.c. line. The stators of both synchros have their leads connected  $S_1$  to  $S_2$ ,  $S_2$  to  $S_3$  and  $S_3$  to  $S_1$ , so the voltage in each of the transmitter stator coils opposes the voltage in the corresponding coils of the receiver. As can be seen a synchro consists of a single phase rotor magnetically coupled to a 3-phase Y-connected stator. When an a.c voltage is applied to the rotor, voltage is induced in each of the stator phases. These induced voltages vary directly, as the sine of the angle between the actual rotor position and a zero reference position. When both transmitter and receiver rotors in a synchros system are on zero or displaced from zero by the same angle, a condition known as 'correspondence' exists. The angle through which a transmitter rotor is mechanically rotated is called a signal. The rotor of a synchro transmitter is usually geared to a manual or mechanical input when the transmitter rotor is turned to a certain angular position  $\Theta^0$ , the rotor of the receiver turns through the same angular position  $\Theta^0$  in synchronism.

✓ Transfer to page 44 under

Fig. 3.3

speed

### 3.4 Improvement of performance of speed/position control systems

The response of most speed/position control systems execute continuous sinusoidal oscillations and may be termed unstable for most practical applications. The performance of the systems can be improved by damping. Damping is a force or signal that opposes the change (or rate of change) of the controlled variable so as to maintain the controlled variable at the desired value or level. The following methods are used for the improvement of their performances.

1. Viscous damping
2. Negative velocity feedback
3. Positive acceleration feedback.
4. Derivative action, and
5. Integral action.

#### 3.4.1 Viscous damping

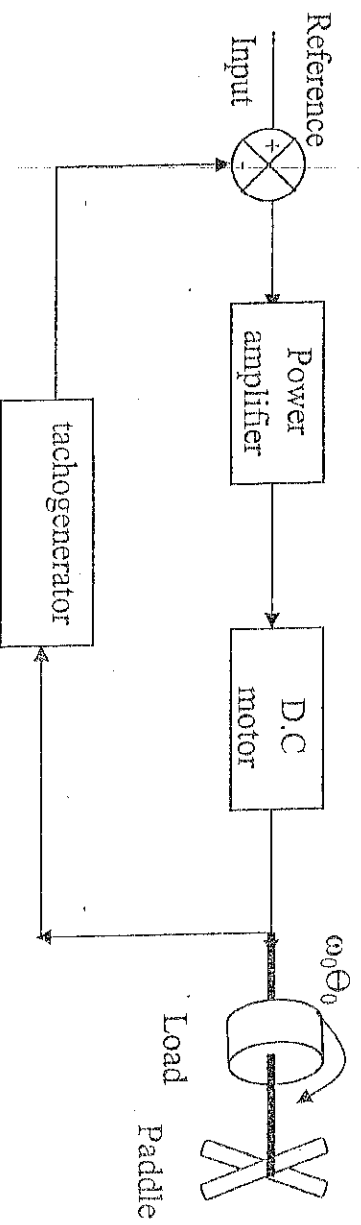


Fig. 3.4.1 Proportional control with viscous damping

Fig. 3.4.1 shows an arrangement how viscous damping is achieved with light paddle. It operates on the fact that frictional forces always oppose motion. To realize this, a light paddle attached to the shaft of the motor carrying the load is immersed in a

viscous fluid and rotates simultaneously with the load whose speed or angular position is being controlled. The paddle helps to minimize the oscillations of the output angular position or speed over the desired value as experienced in the case without damping. The feedback circuit containing the tachogenerator provides the normal proportional control that checks the deviation of the output from the desired or set (or reference) value.

### 3.4.2 Negative velocity feedback

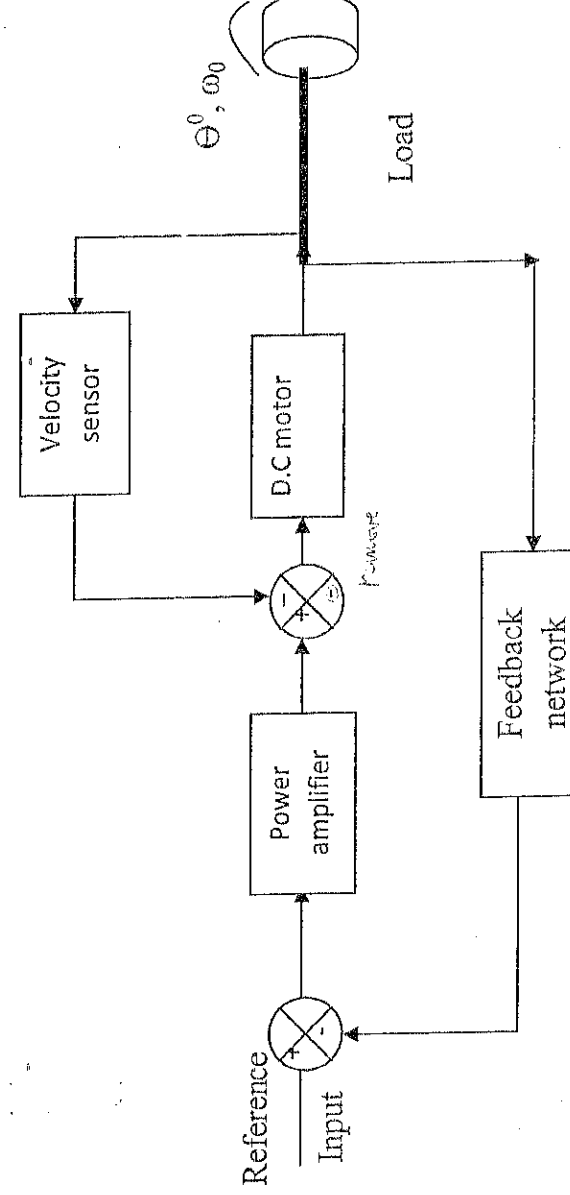


Fig. 3.4.2 Proportional control with negative velocity damping

Fig. 3.4.2 illustrates the principle operation of proportional control with negative velocity damping. Velocity feedback is widely used to stabilize a system which tends to oscillate. If the output of a system is angular motion  $\Theta_0$ , then the rate of change,  $\frac{d\Theta_0}{dt}$  is a true velocity. When a sudden (step) change is made, the error is maximum and so the output changes very rapidly. The velocity is then greatest at the

angular  
of the  
the case  
desires the  
desired

start which is fed back. The effects of the feedback is to reduce the error in a manner directly proportional to the velocity. The feedback increases the stiffness of the shaft thereby providing damping which is useful in stabilizing and oscillatory system. The feedback path containing the feedback network performs the conventional proportional control action that drives the output to the desired value.

### 3.4.3 Positive acceleration feedback

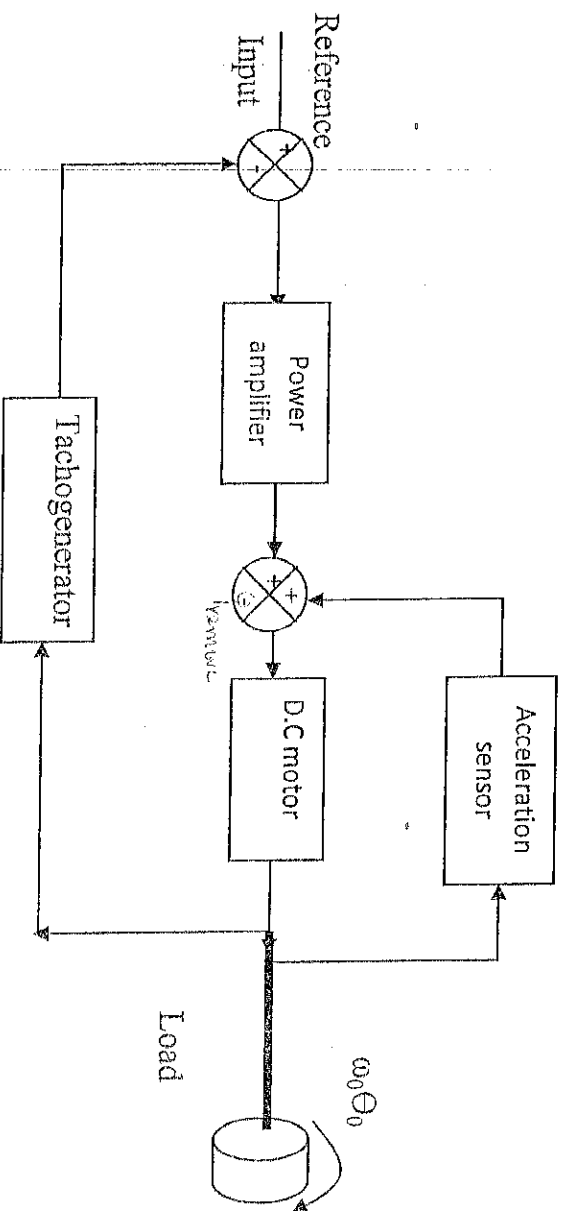
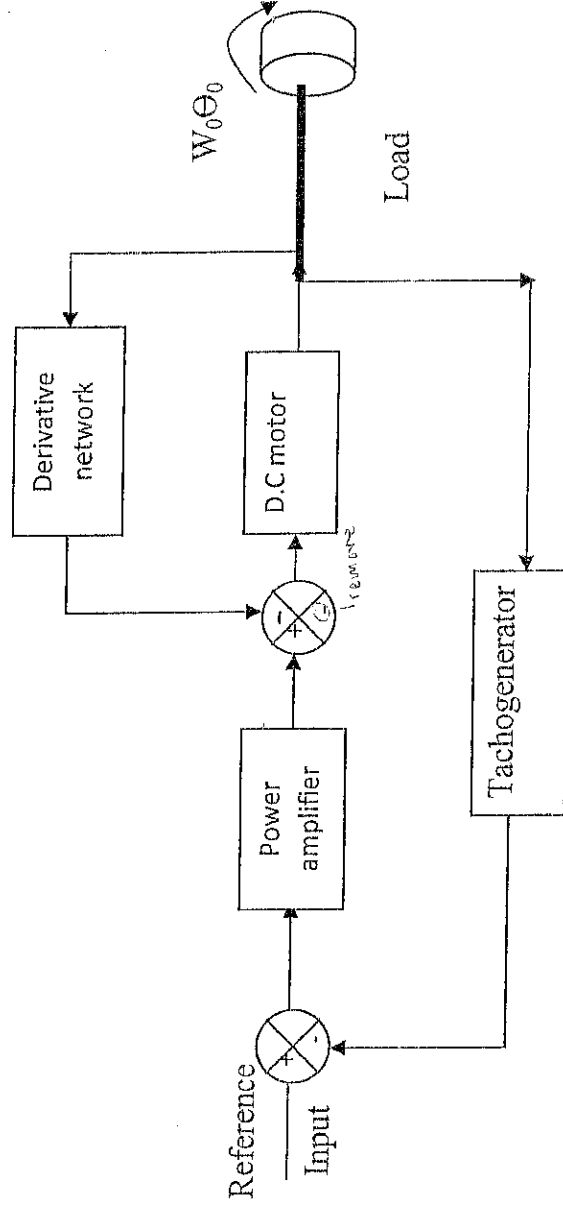


Fig. 3.4.3 Proportional control with positive acceleration feedback

Fig. 3.4.3 shows an arrangement illustrating the working principle of proportional control with positive acceleration feedback. Positive acceleration feedback adds 'electronic' inertia to the system. The 'electronic' inertia makes disturbance feel as if it is acting on a large mass which means the stiffness of the system has increased. In other words, a system with positive acceleration feedback will have less change when experiencing the same disturbances unlike a system without positive acceleration feedback. Introducing positive acceleration feedback loop has the effect of increasing the effective mass, inertia, stiffness, and damping of the system which protects it from oscillations thereby improving the performance of the system. The

feedback loop containing the tachogenerator performs the conventional proportional action that maintains the output of the system at the desired value.

#### 3.4.4 Proportional plus derivative (PD) action

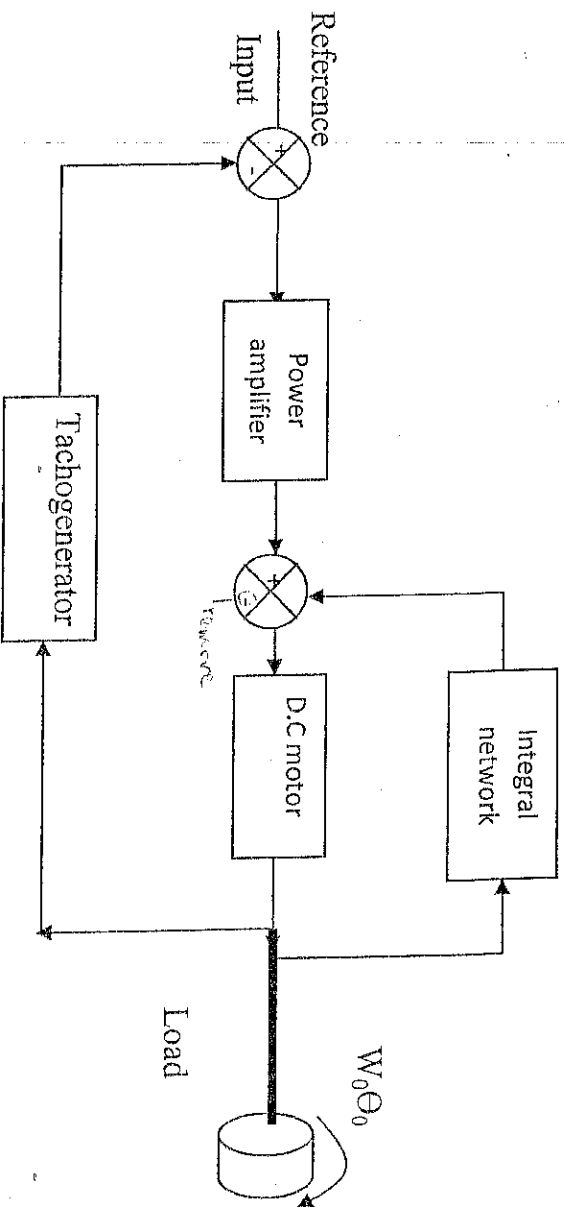
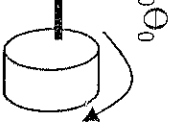


*Fig. 3.4.4 proportional plus derivative control system*

Fig. 3.4.4 illustrates the principle of operation of the proportional plus derivative control system. This system employs the proportional control together with the derivative action. The feedback loop containing the tachogenerator network performs the proportional action. Proportional action reacts to the size of the deviation from the set point (desired value) when sending a corrective signal. The size of the corrective signal can be adjusted in relation to the size of the error by reducing its value gradually to zero. On the other hand, the derivative action reacts to speed in which the deviation is changing. The corrective signal will be proportional to the rate of change within the system which provides the damping action. The feedback loop containing the derivative network performs this action and fed back in the manner shown.



### 3.4.5 Proportional plus integral (PI) action



*Fig. 3.4.5 Proportional plus integral control system*

Fig. 3.4.5 is the block diagram of a proportional plus integral control system. The feedback loop containing the tachogenerator performs the proportional action by reacting to the size of the deviation from the set point when sending a corrective signal. The size of the corrective signal can be adjusted in relation to the size of the error by reducing its value gradually to zero.

On the other hand, the feedback loop containing the integral network performs the integral action. This action reacts to the length of the time that the deviation from the set point exists when sending a corrective signal. The longer the error exists the greater the corrective signal. The longer the error exists the greater the corrective signal. This action applies ever-increasing effort until the error is reduced to zero. If the correction being applied does not bring the process value output up to set point, for whatever reason, integral action increasingly moves the proportional band relative to the set point until the process value error is reduced to zero and the set point is achieved.

### 3.4.6 Proportional plus integral and directive (PID) action

The block diagram of proportional plus integral and directive controller is in figure 3.4.6

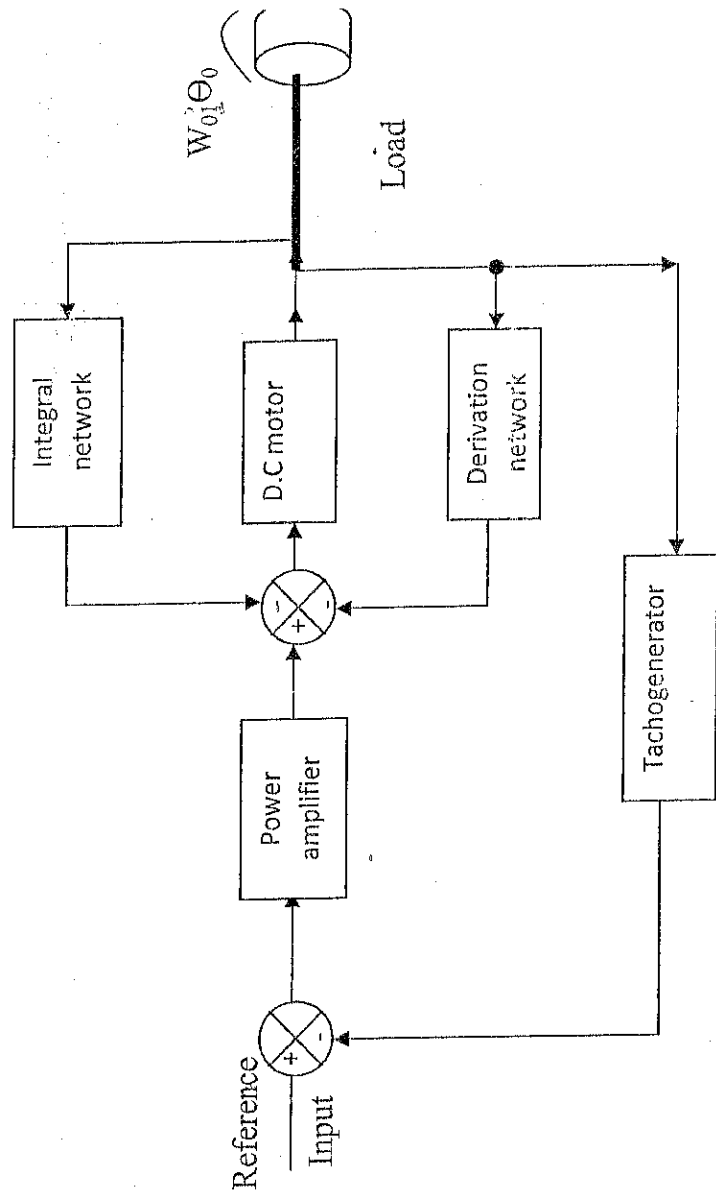


Fig. Proportional plus integral and derivative action

#### Example

If the input and feedback to potentiometers of closed loop position control system have the same constant  $K_P = 100\text{volts/rad}$ , and  $\Theta_i$  and  $\Theta_0$  are input and output angular positions respectively, calculate the error voltage and explain the action of the drive motor if,

- (a)  $\Theta_i = 0.5\text{ rad}$  and  $\Theta_0 = 0.5\text{rad}$
- (b)  $\Theta_i = 0.6\text{ rad}$  and  $\Theta_0 = 0.5\text{rad}$
- (c)  $\Theta_i = 0.6\text{rad}$  and  $\Theta_0 = 0.7\text{rad}$